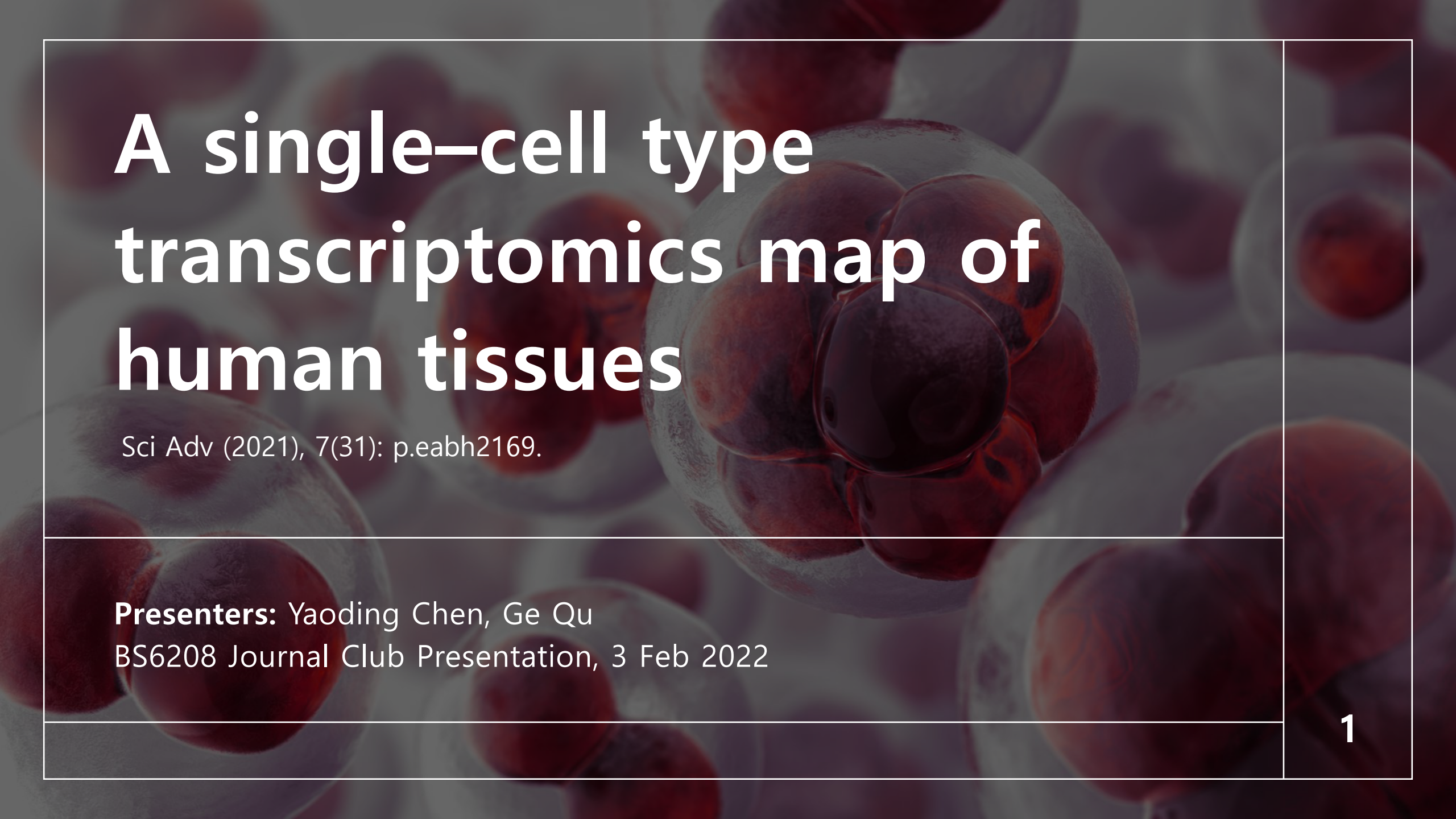




A single-cell type transcriptomics map of human tissues

Sci Adv (2021), 7(31): p.eabh2169.

Presenters: Yaoding Chen, Ge Qu
BS6208 Journal Club Presentation, 3 Feb 2022

Outline

Introduction & method

- What does single cell sequencing do?
- What is a cell type? How does it link to biological function?

Results

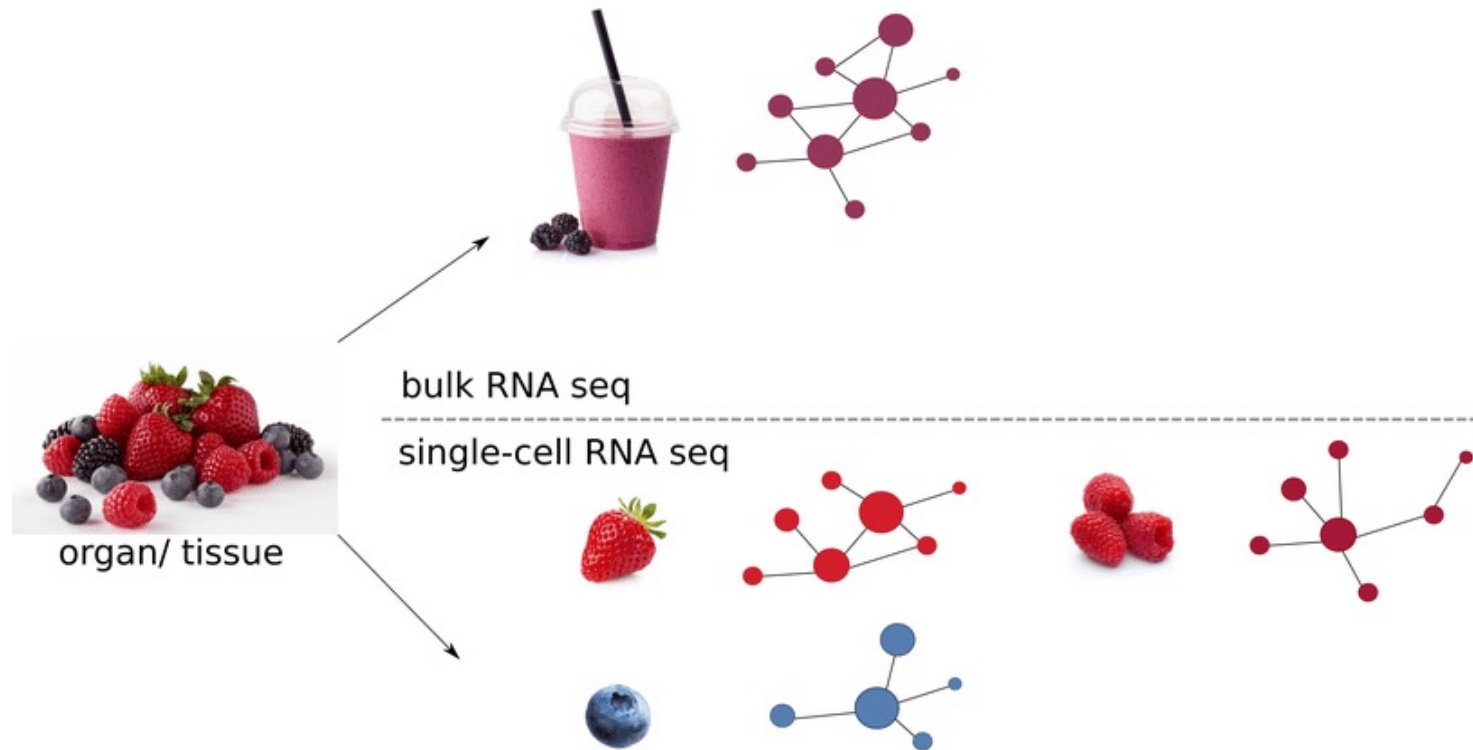
- Do cells with similar functions have similar gene profiles?
- Are tissues or organs just a mixture of different types of cells in different proportion?

Discussion

- Do we have a clear answer yet?

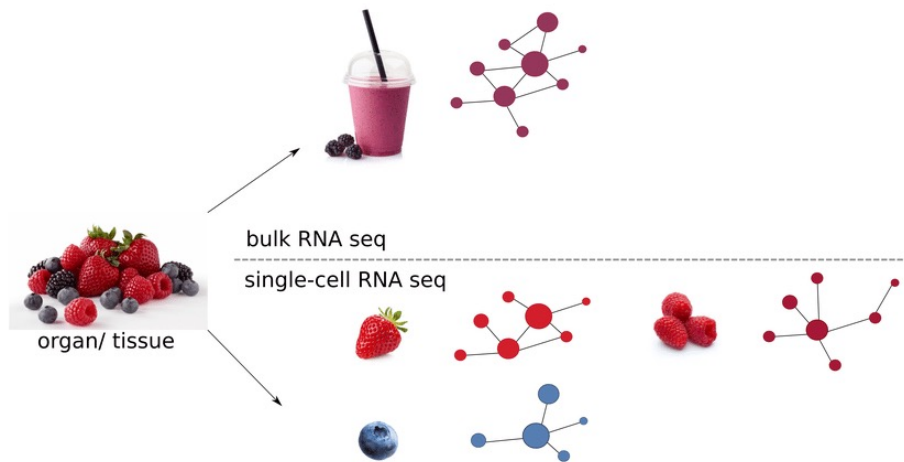
Introduction:

single cell technique reveal ingredients of life



Introduction:

two angles for single cell technique applications



**How can I preserve the original milkshake?
Or what exactly goes wrong in pathogenesis?**

Which organ > which kind of cells?
Patient variation in drug response explained?

.....

**How can I make a new or better milkshake?
Or can we regenerate similarly healthy cells?**

In model animals for drug screening?
In human for replacement therapies?

.....

Introduction:

what is a cell type?



Original
not available



Which to choose?

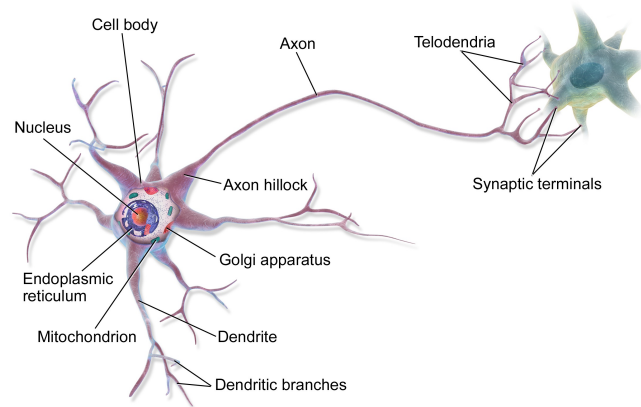
Not all
syrup is for
milkshakes



- Different syrup, still the same milkshake?
- Do cells of the same cell type have similar features?

Introduction:

what is a cell type?



- Definition of syrup is based on,
 - **Functional elements:** sugar?
 - **Morphology:** a thick, viscous liquid
 - **Locations:** typically in kitchen, supermarket, warehouse...
- Cell type classification is typically based on biological function, which are executed by certain proteins.
- For example, for a neuron cell:
 - **Function elements:** certain proteins
 - **Morphology:** octopus-like
 - **Locations:** mostly in the nervous system

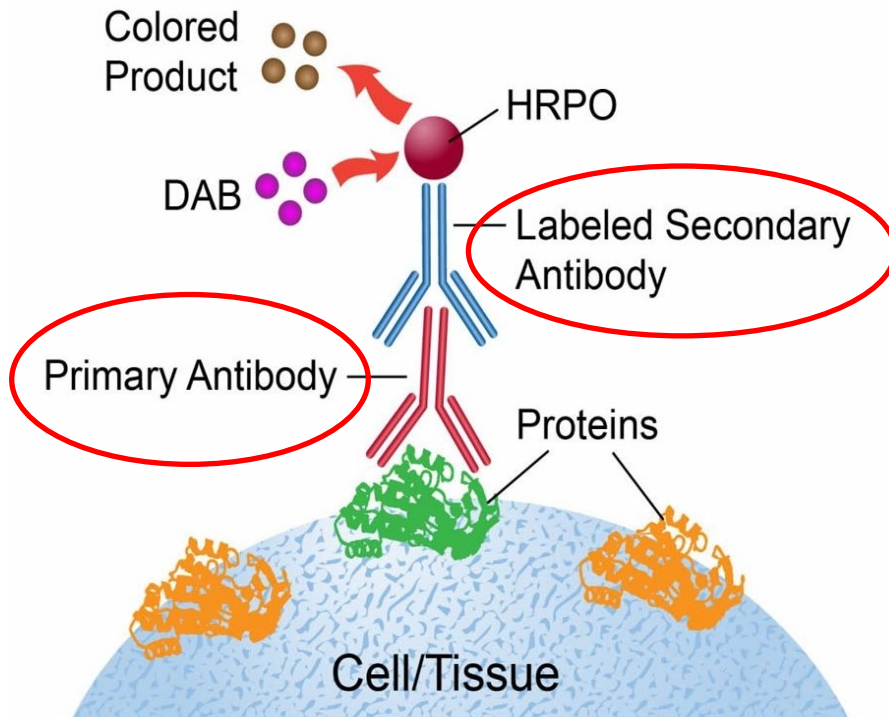
Methods:

How we identify cell types?

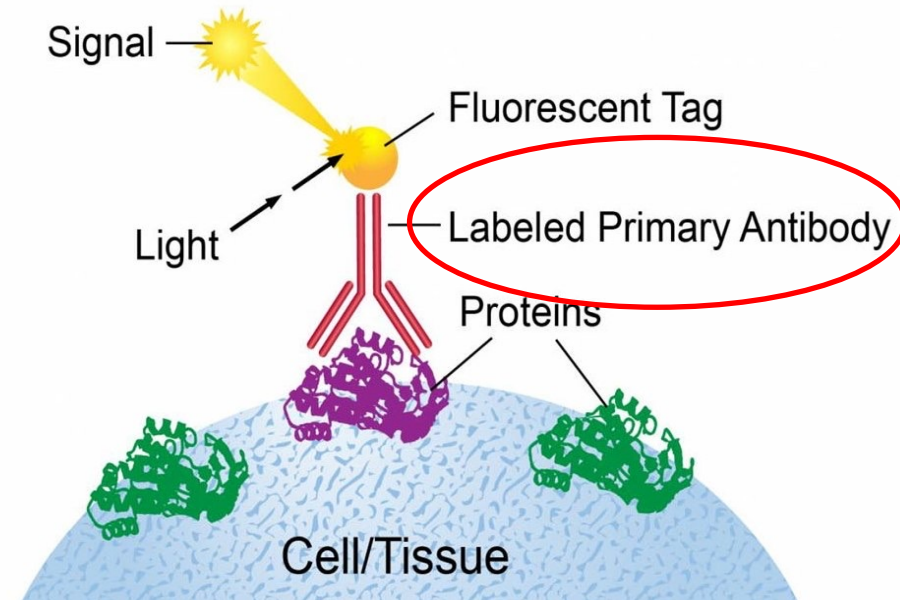
Antibody tagging is the key!

Antibody is protein-specific binding proteins

Indirect Immunohistochemistry



Immunofluorescence



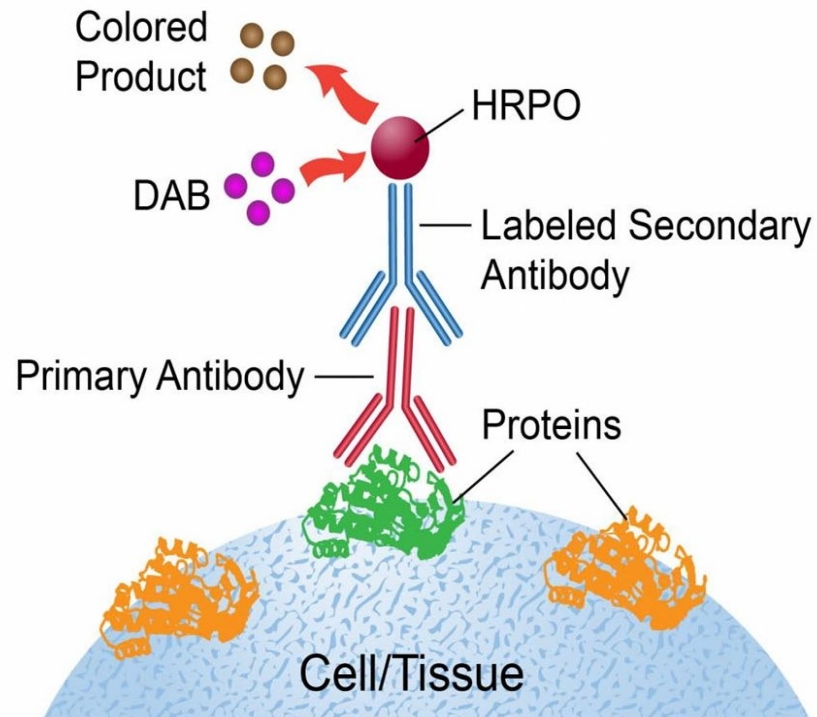
Methods:

How we identify cell types?

Antibody tagging is the key!

Antibody is hard to design and manufacture!

Indirect Immunohistochemistry



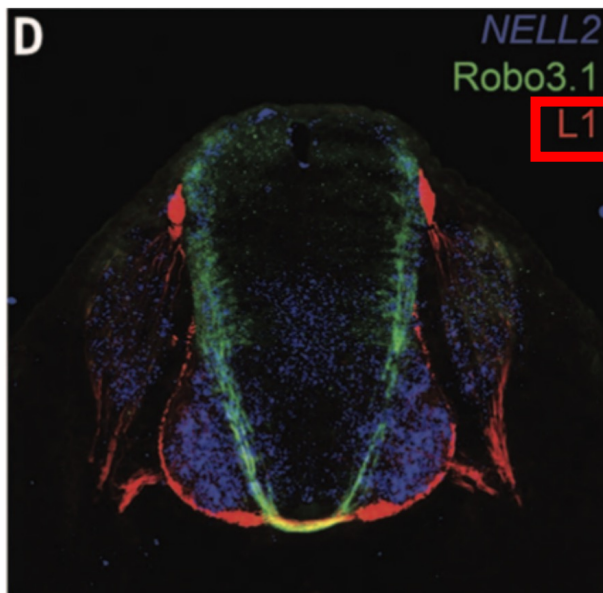
Prof. Mathias Uhlén, antibody expert @ KTH

*The correspondent author of the paper

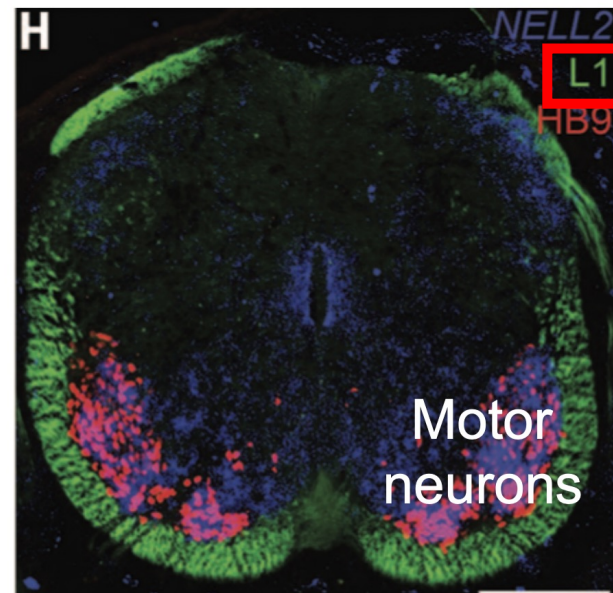
Protein tagging made easy with his work!

Introduction:

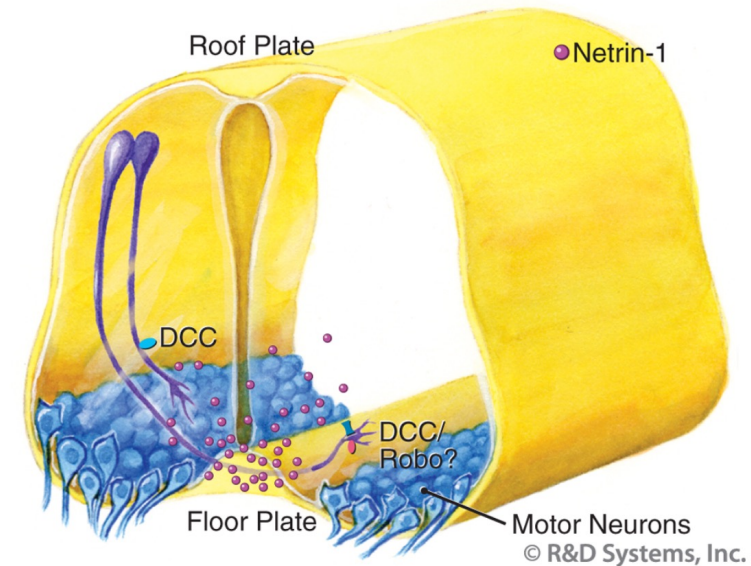
Are we always right about cell types?



Mouse Embryonic Day 10.5



Mouse Embryonic Day 12.5



The same cell type, different location, different time, different protein >> different function

Methods:

A missing link between function & location



Syrup

Cell type

Defined by 3+
biomarkers & location

Chemical
profile

Gene expression
profile

- Can we infer cell types or function from RNA profiles?
- What contributes to cell heterogeneity?



Sugar

Protein X

Related to certain
biological function

Methods:

A missing link between function & location

Article | [Open Access](#) | [Published: 09 October 2020](#)

Single-cell and spatial transcriptomics enables probabilistic inference of cell type topography

[Alma Andersson](#) , [Joseph Bergenstråhle](#), [Michaela Asp](#), [Ludvig Bergenstråhle](#), [Aleksandra Jurek](#), [José Fernández Navarro](#) & [Joakim Lundeberg](#) 

[Communications Biology](#) **3**, Article number: 565 (2020) | [Cite this article](#)

18k Accesses | **46** Citations | **4** Altmetric | [Metrics](#)

The earlier work by the lab suggest a knowledge gap between spatial information and cell type, despite available single cell and spatial RNAseq data

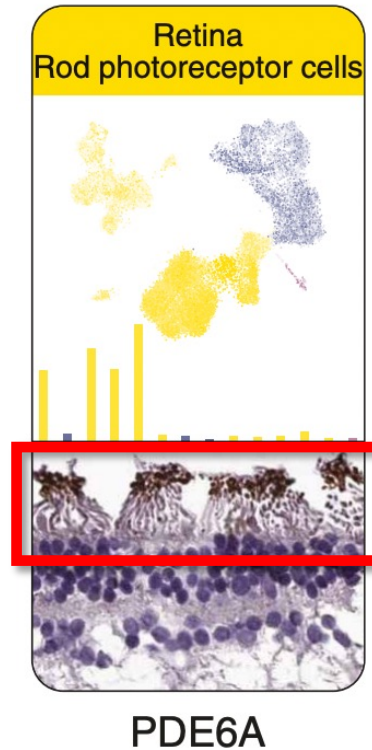
Methods:

A missing link between function & location

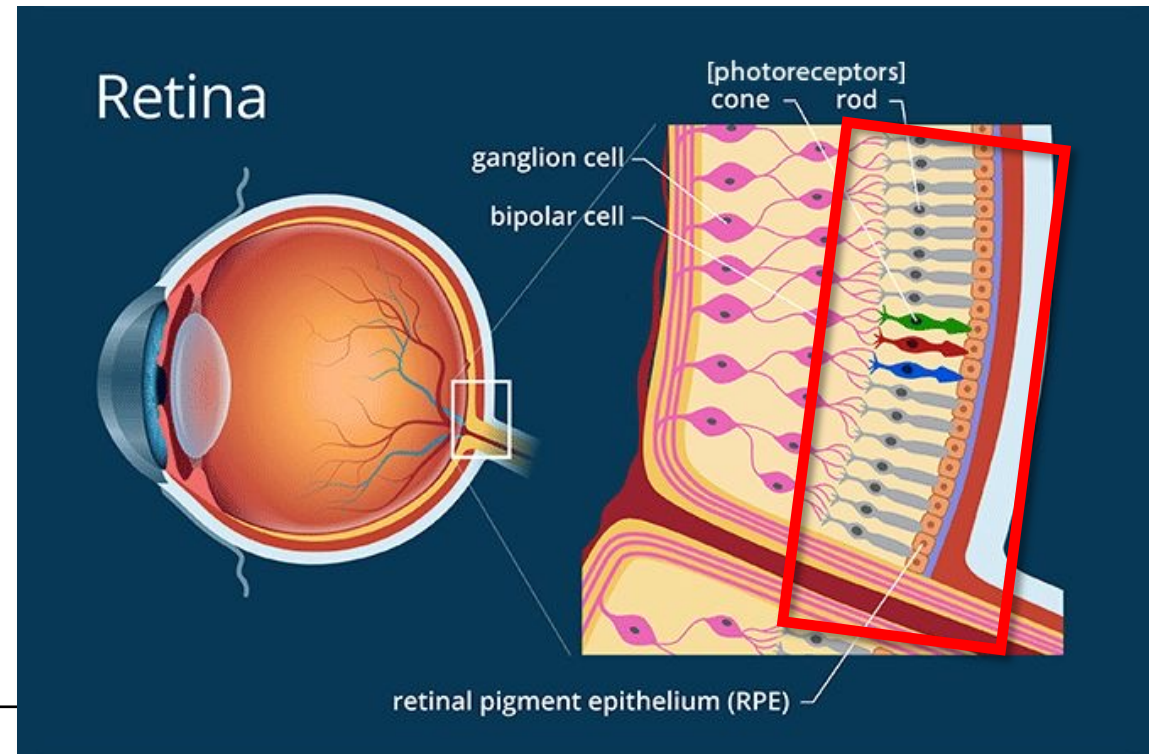
Cell type

Proteomics

Protein X
(function)



Antibody tagging indicates fine anatomy
location of the cell type



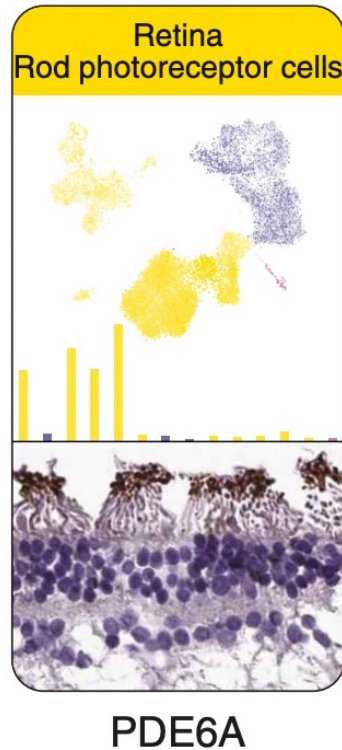
Methods:

Data sources and dataset selection

Cell type

Proteomics

Protein X
(function)



Data sources

scRNAseq data from
Human Cell Atlas

bioimaging data from
Human Protein Atlas

51 cell types

13 tissues+ Blood

Exceptions: eye, lung,
pancreas & rectum due to
data availability

Selection criteria

10XGenomics-based
scRNAseq only

> 4k cells
> 4M read counts

No pre-enrichment
of cell types

Expression profiles
highly correlated with
tissue profile

Methods:

Measuring cell type similarity



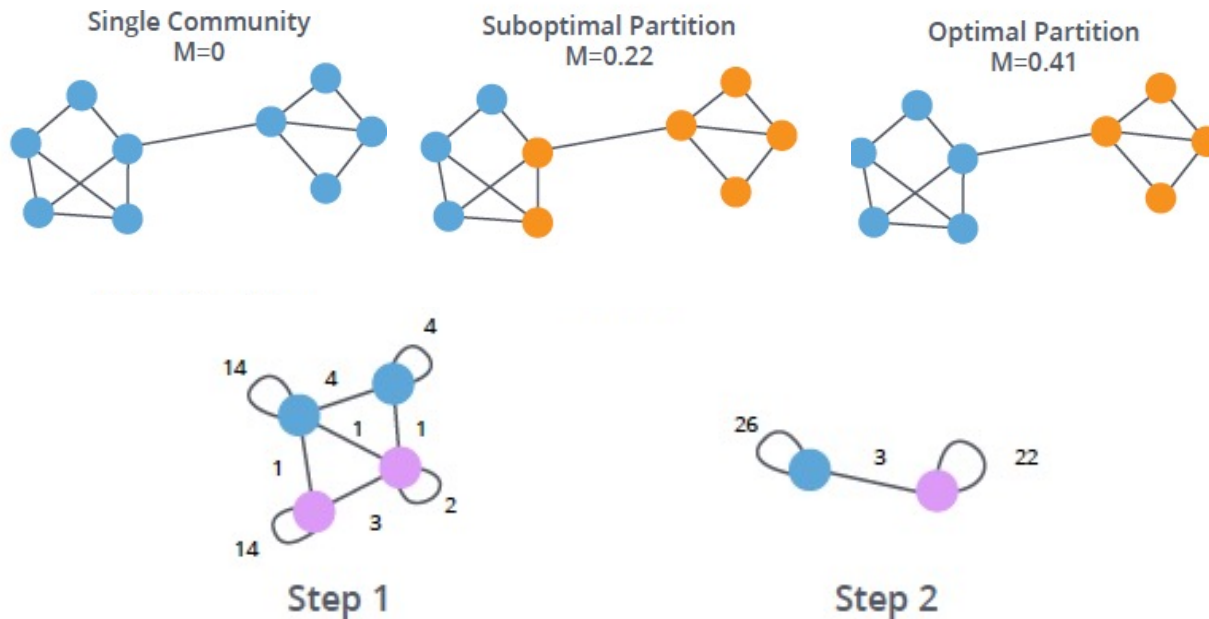
Clustering algorithm:

Louvain method for community detection
(based on KNN graph)

1. Initialise a network where cells are nodes, cell-cell distance are weight
2. Detect local communities to maximise modularity (M)
3. Make local communities a node of a new network to maximise M
4. Repeat Step 2&3 until maximal M

Advantages of graph-based clustering:

- Works well in large datasets due to scalability and speed
- No need to define cluster number



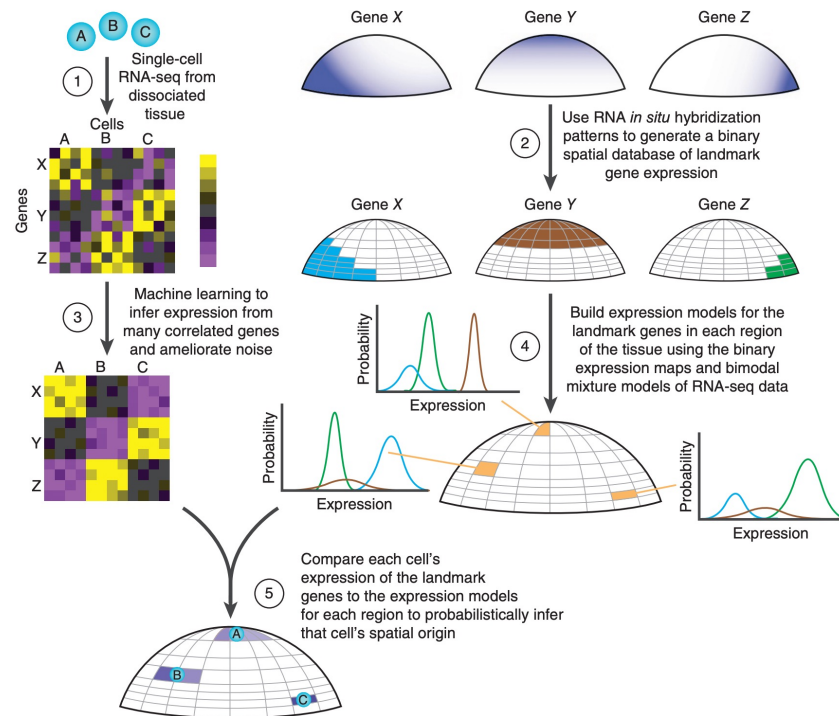
Steps 1 and 2 repeat in passes until there is no further increase in modularity or a set number of iterations have occurred.

Methods:

Measuring cell type similarity



Another View of Modularity



normalization

adjacency matrix

probability a random edge would go between i and j

$$Q = \frac{1}{4m} \sum_{i,j \text{ in same module}} \left(A_{ij} - \frac{k_i k_j}{2m} \right)$$

$m = \# \text{ edges in graph}$
 $k_i = \text{degree}(i)$

Consider the case of only 2 modules.

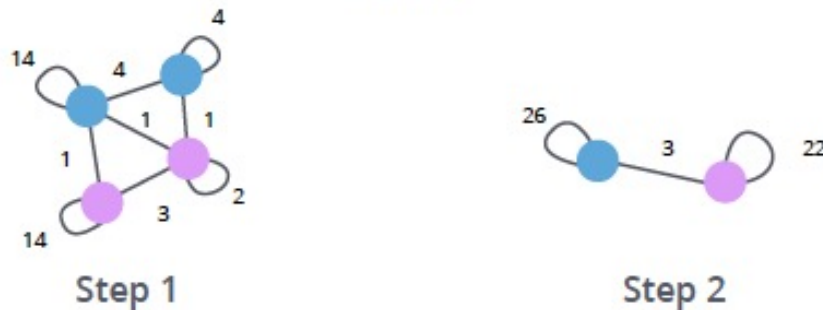
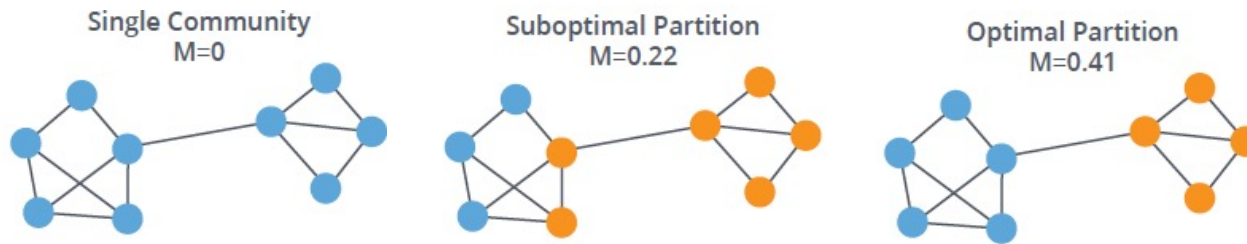
Let $s_i = 1$ if node i is in module 1; -1 if node i is in module 2

$$Q = \frac{1}{4m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) (s_i s_j + 1)$$

$$= \frac{1}{4m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) s_i s_j$$

Methods:

Measuring cell type similarity



Steps 1 and 2 repeat in passes until there is no further increase in modularity or a set number of iterations have occurred.

Clustering algorithm:

Louvain method for community detection (based on KNN-graph)

- Distance gets similar due to the law of large numbers, making estimation inaccurate
- Distortion when highly dense in a KNN graph
- No objective way to determine optimise parameters

Methods:

Measuring cell type similarity



Name	Year	Method type	Strengths	Limitations
scanpy ⁴	2018	PCA + graph-based	Very scalable	May not be accurate for small data sets
Seurat (latest) ³	2016			
PhenoGraph ³²	2015			
SC3 (REF. ²²)	2017	PCA + k-means	High accuracy through consensus, provides estimation of k	High complexity, not scalable
SIMLR ²⁴	2017	Data-driven dimensionality reduction + k-means	Concurrent training of the distance metric improves sensitivity in noisy data sets	Adjusting the distance metric to make cells fit the clusters may artificially inflate quality measures
CIDR ²⁵	2017	PCA + hierarchical	Implicitly imputes dropouts when calculating distances	
GiniClust ⁷⁵	2016	DBSCAN	Sensitive to rare cell types	Not effective for the detection of large clusters
pcaReduce ²⁷	2016	PCA + k-means + hierarchical	Provides hierarchy of solutions	Very stochastic, does not provide a stable result
Tasic et al. ²⁸	2016	PCA + hierarchical	Cross validation used to perform fuzzy clustering	High complexity, no software package available
TSCAN ⁴¹	2016	PCA + Gaussian mixture model	Combines clustering and pseudotime analysis	Assumes clusters follow multivariate normal distribution
mpath ⁴⁵	2016	Hierarchical	Combines clustering and pseudotime analysis	Uses empirically defined thresholds and a priori knowledge
BackSPIN ²⁶	2015	Biclustering (hierarchical)	Multiple rounds of feature selection improve clustering resolution	Tends to over-partition the data
RaceID ²³ , RaceID2 (REF. ¹¹⁵), RaceID3	2015	k-Means	Detects rare cell types, provides estimation of k	Performs poorly when there are no rare cell types
SINCERA ⁵	2015	Hierarchical	Method is intuitively easy to understand	Simple hierarchical clustering is used, may not be appropriate for very noisy data
SNN-Cliq ⁸⁰	2015	Graph-based	Provides estimation of k	High complexity, not scalable

Clustering algorithm:
Louvain method for community detection (based on KNN-graph)

- With the accumulation of the data, pairwise distance will be similar due to the law of large numbers, which makes estimation inaccurate
- No objective way to determine optimise parameters

Outline

Introduction & method

- What does single cell sequencing do?
- What is a cell type? How does it link to biological function?
- How do we study cell types with scRNAseq?

Results

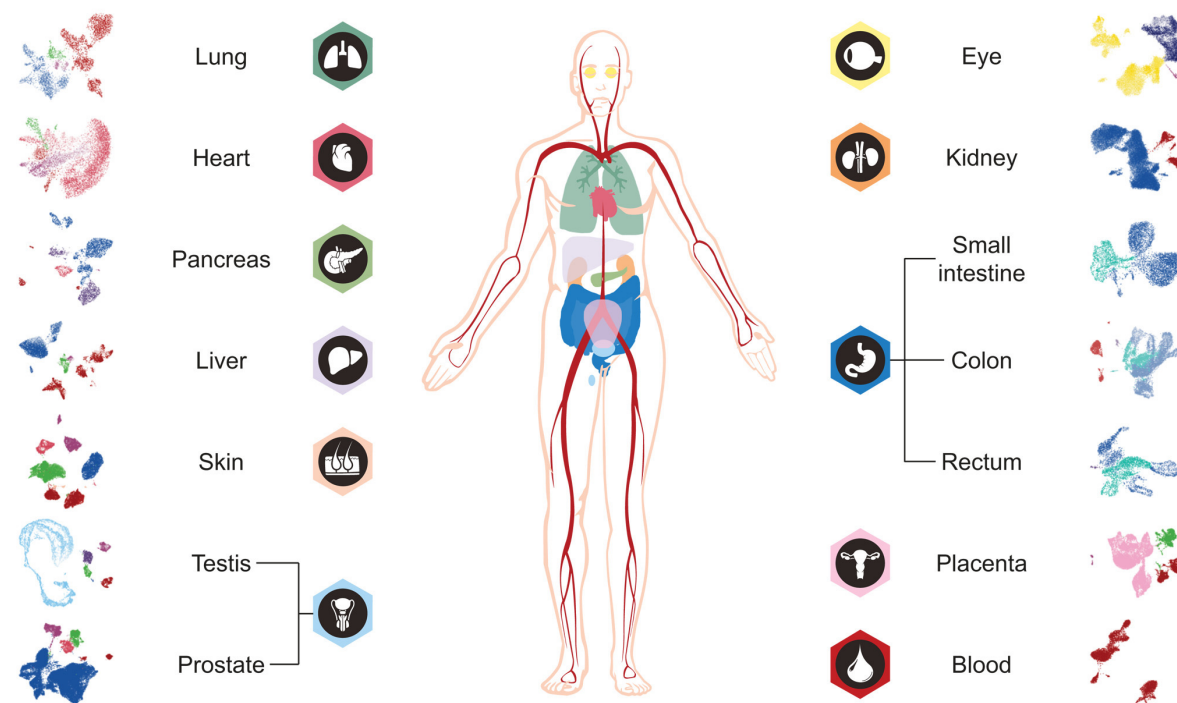
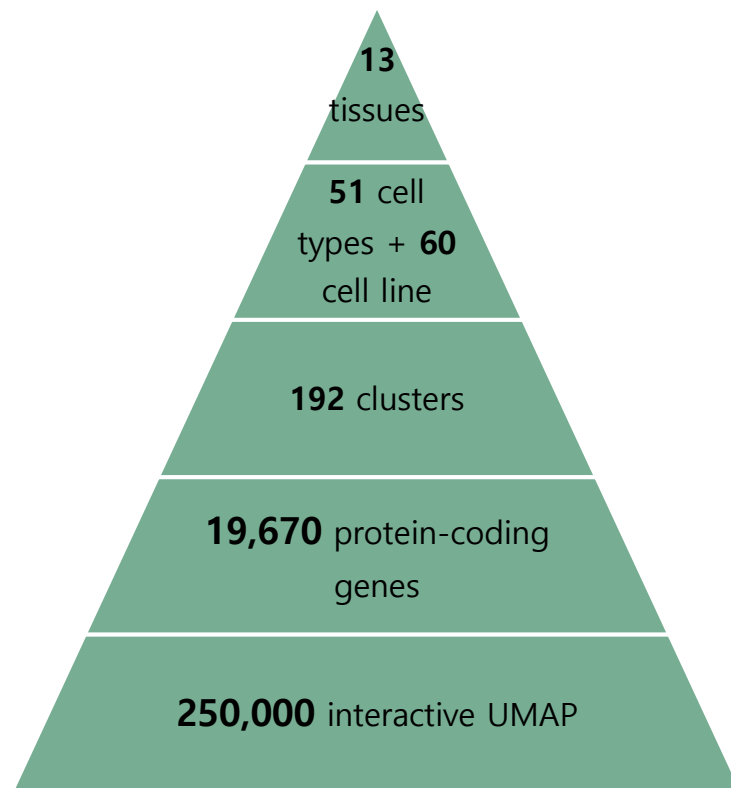
- Do cells with similar functions have similar gene profiles?
- Are tissues or organs just a mixture of different types of cells in different proportion?

Discussion

- Do we have a clear answer yet?

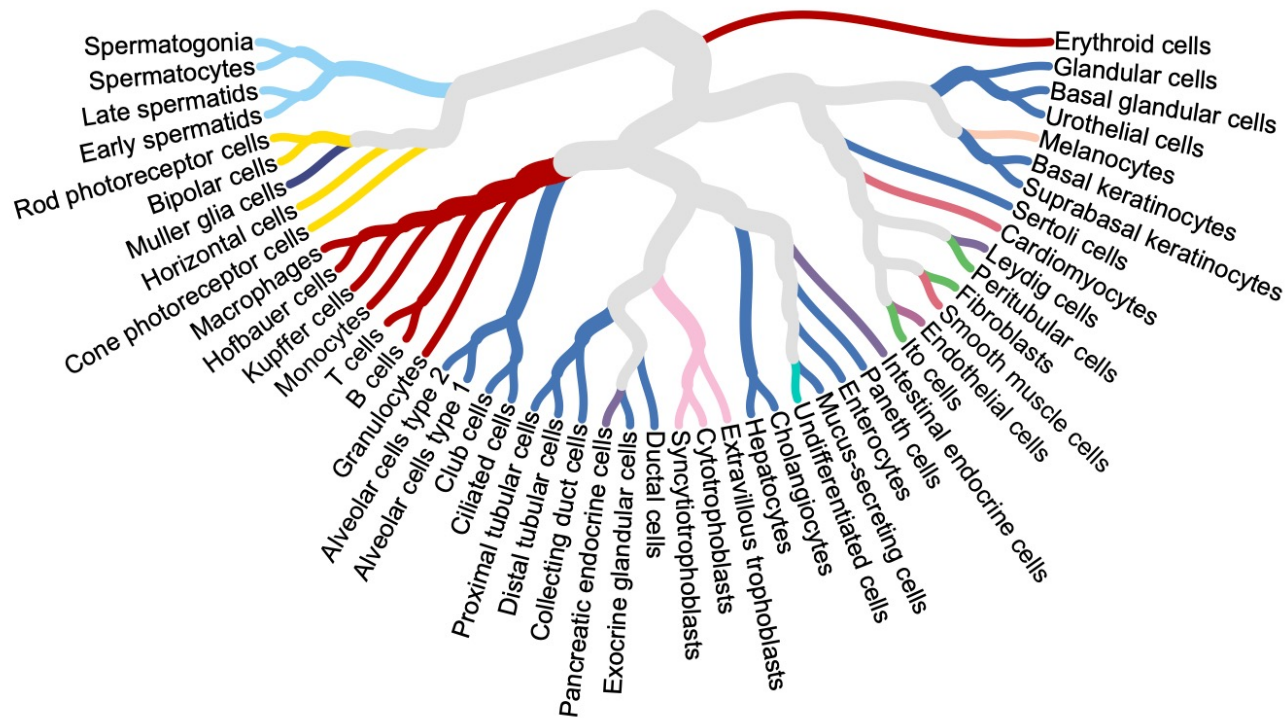
Result summary:

A new knowledge portal

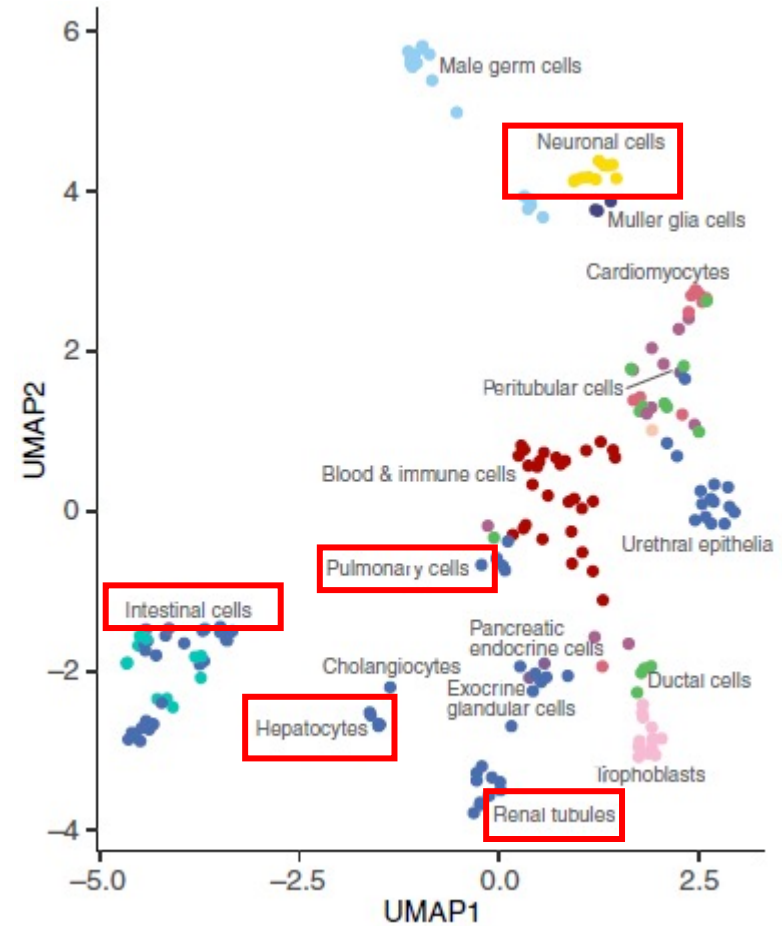


Result:

How similar are the cell types?

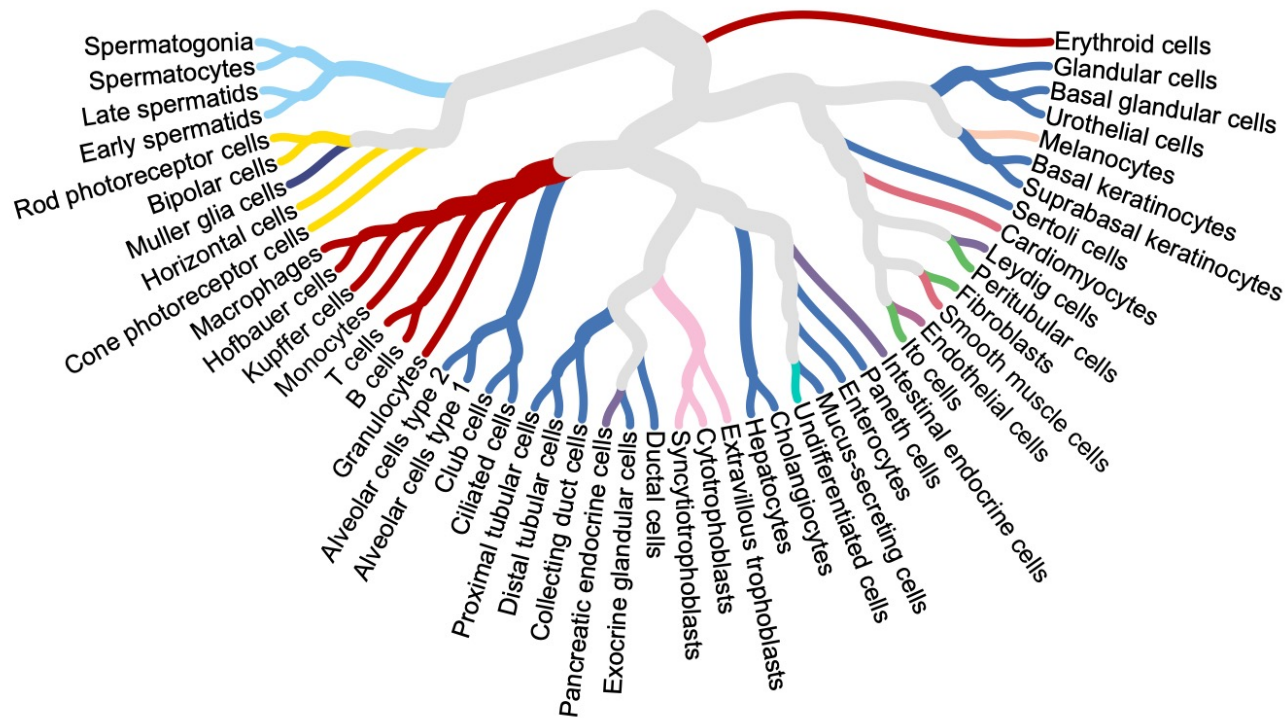


Differences between transcriptomic-based tissue-specific and cell type-specific clustering

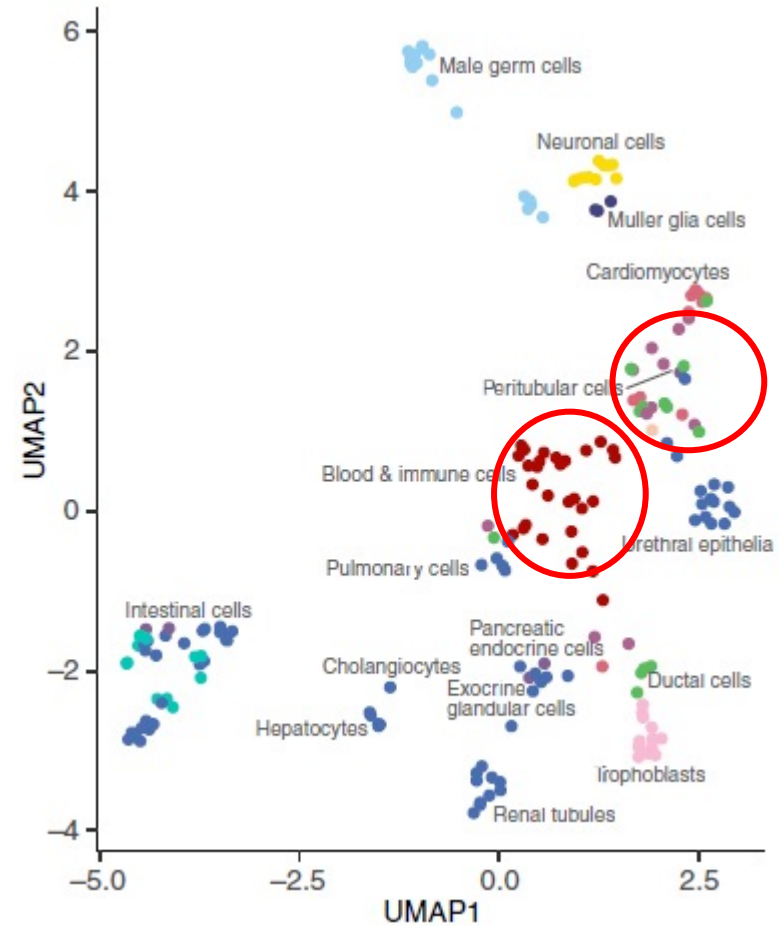


Result:

How similar are the cell types?



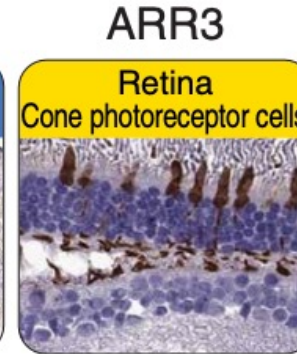
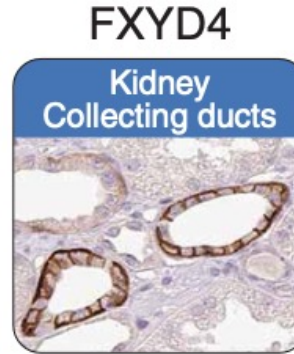
Differences between transcriptomic-based tissue-specific and cell type-specific clustering



Result: Cell type specificity of genes

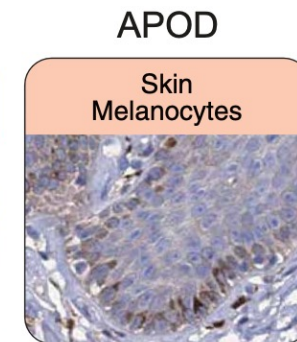
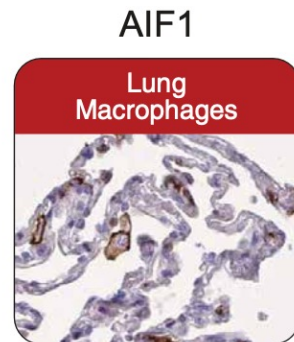
Low cell type specificity
Low variance, housekeeping

Not detected

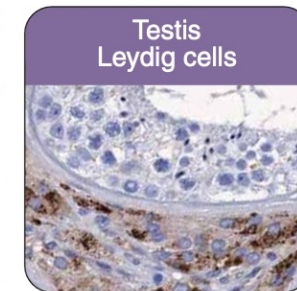
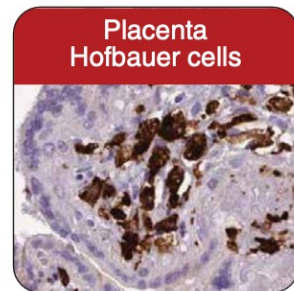


Single **cell type enriched**
(4 X than all)

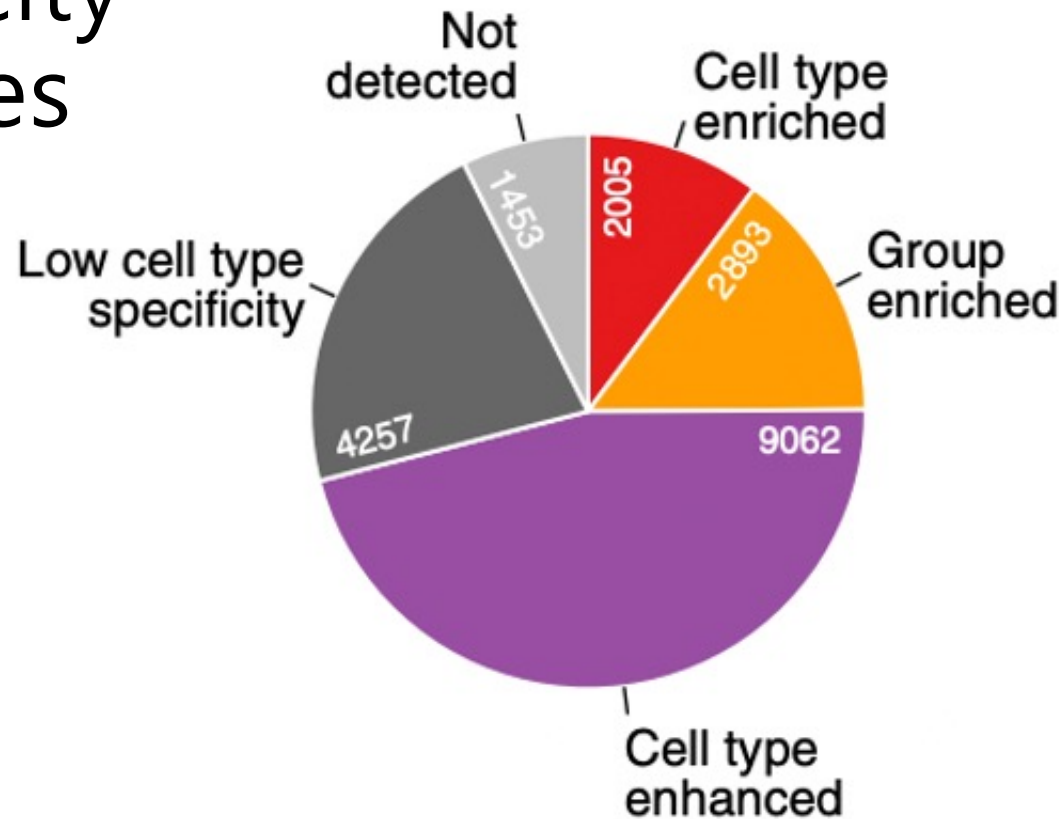
Single **cell type enhanced**
(4 X mean of the rest)



Group enriched
(2~10 4 X than all)



Result: Cell type specificity of genes



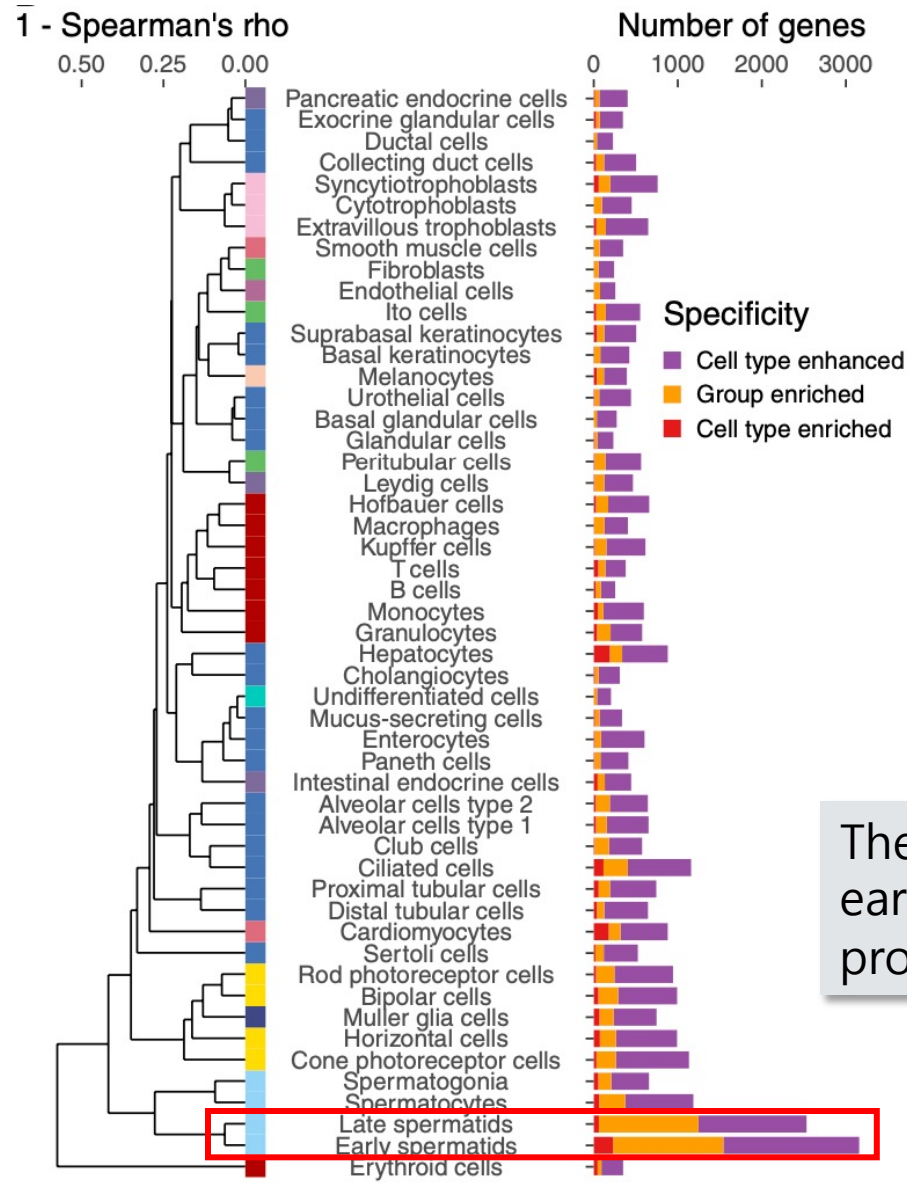
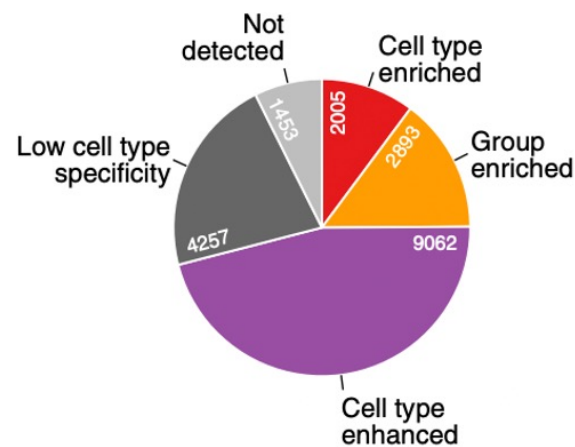
Single **cell type enriched**
(4 X than all)

Single **cell type enhanced**
(4 X mean of the rest)

Group enriched
(2~10 4 X than all)

Low cell type specificity
Low variance

Result: Similar cell type, similar genes?



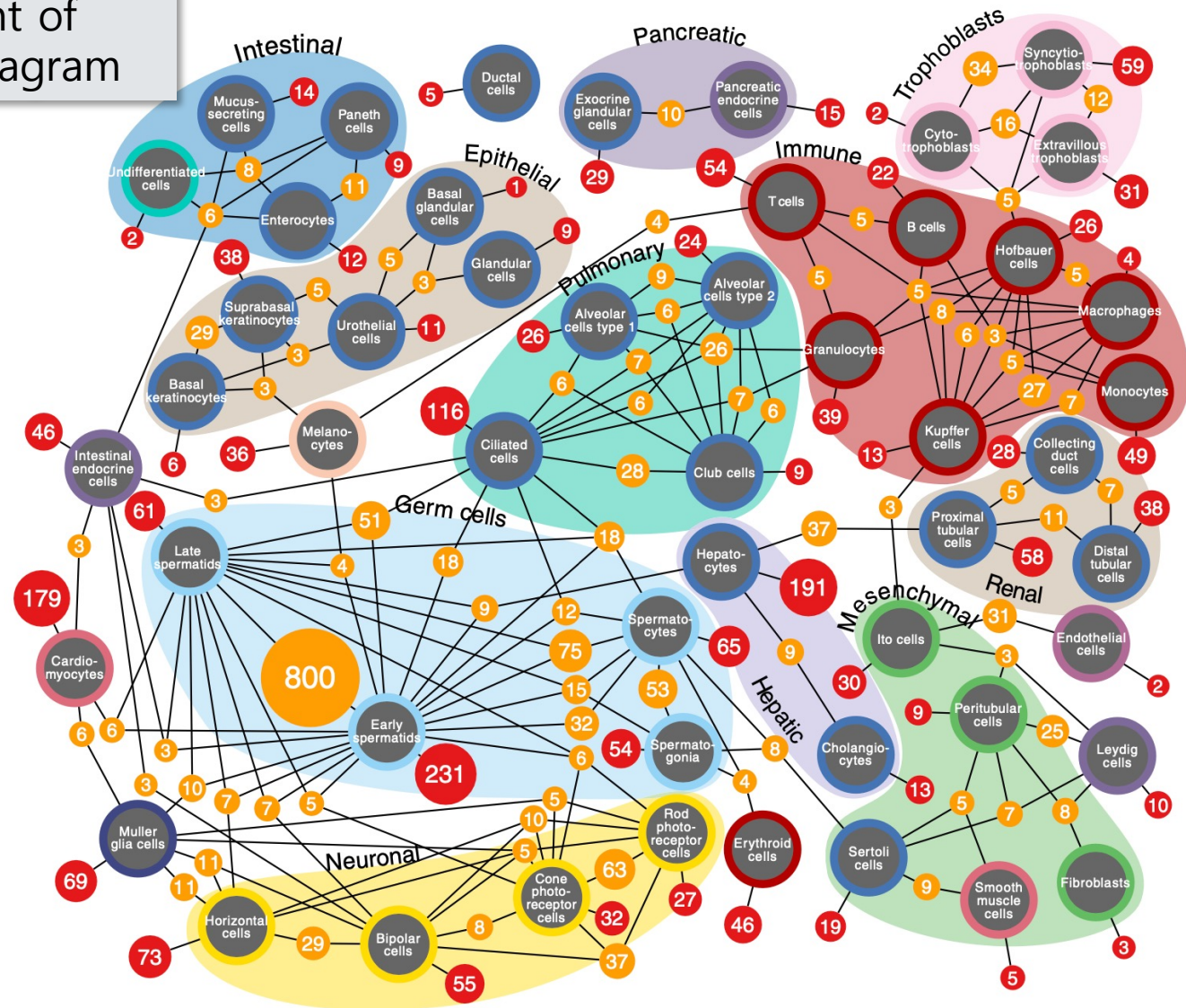
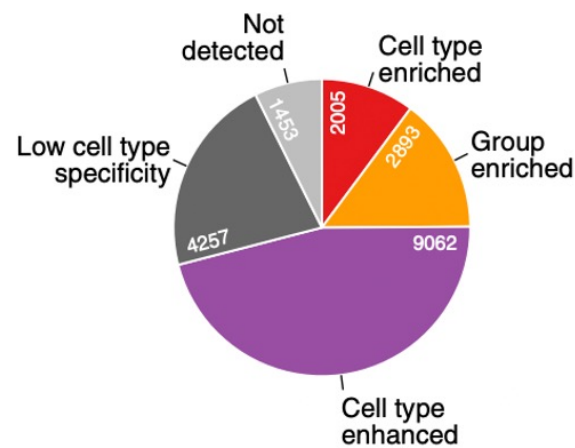
They are linked to key early developmental processes



Result:

Similar cell
type,
similar
genes?

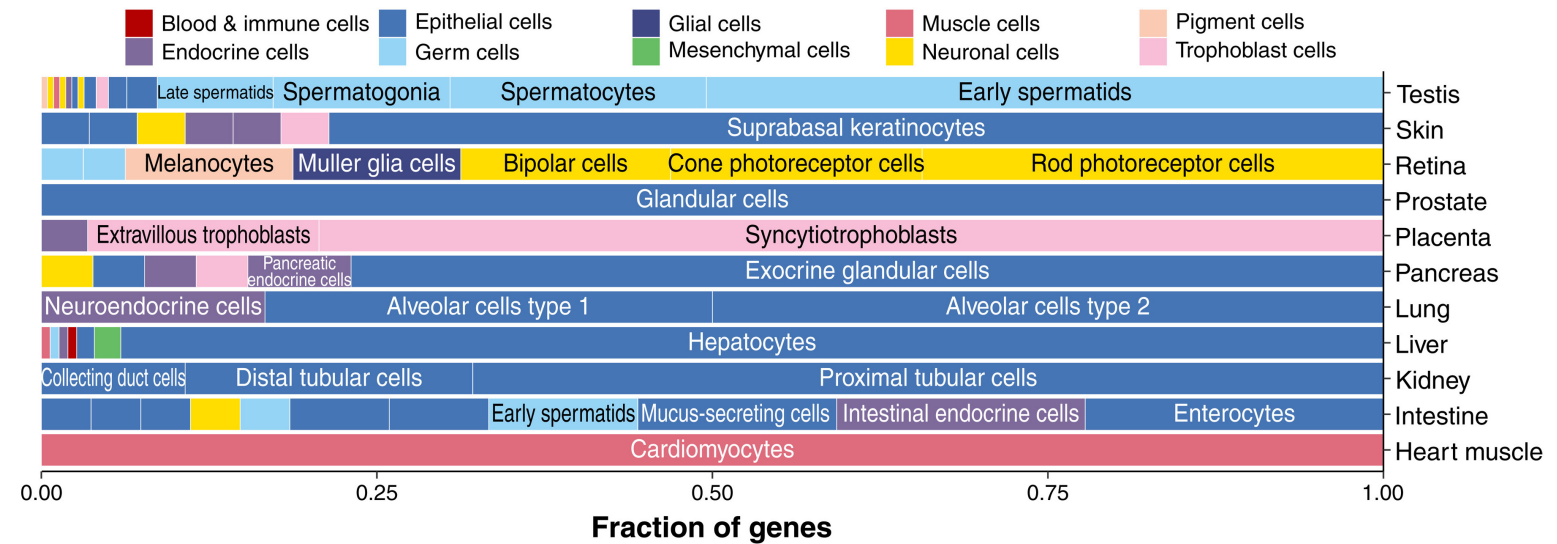
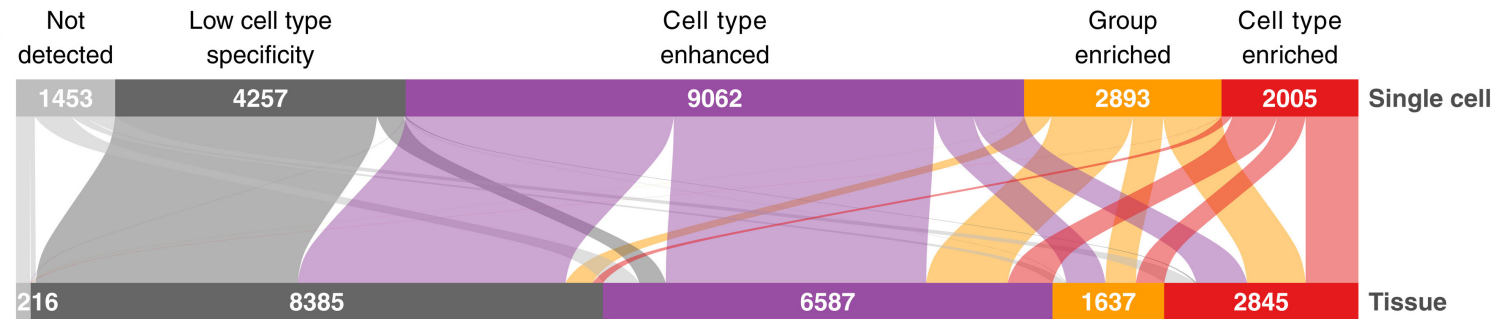
A variant of Venn diagram



- Blood & immune cells ● Germ cells ● Muscle cells ● Trophoblast cells
● Endocrine cells ● Glial cells ● Neuronal cells ● Undifferentiated cells
● Epithelial cells ● Mesenchymal cells ● Pigment cells ● Vascular cells

Result:

Tissue =
Sum of all?

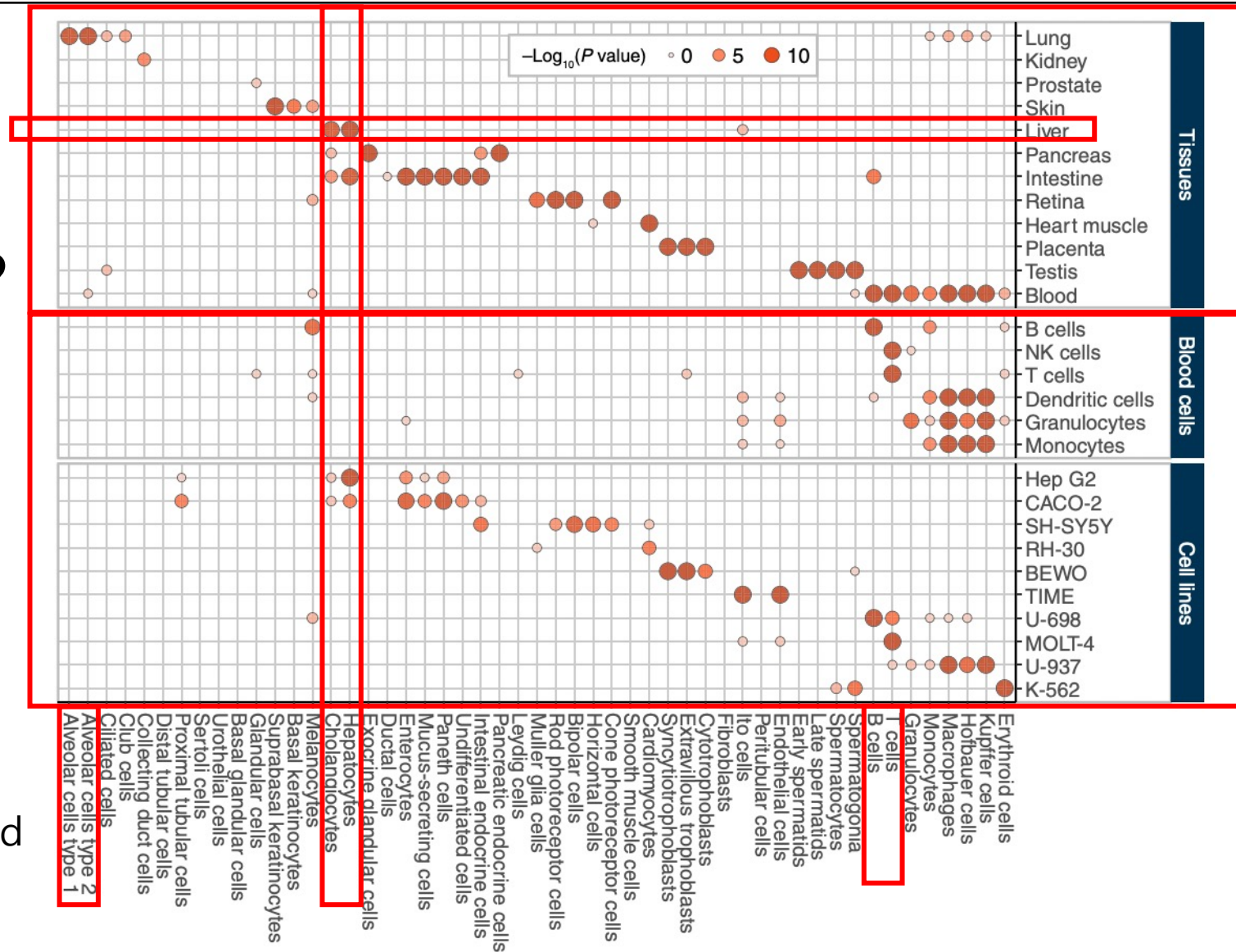


Result: Tissue = Sum of all?

CACO-2 = colon cancer
Hep G2 = derived from liver

Cell line

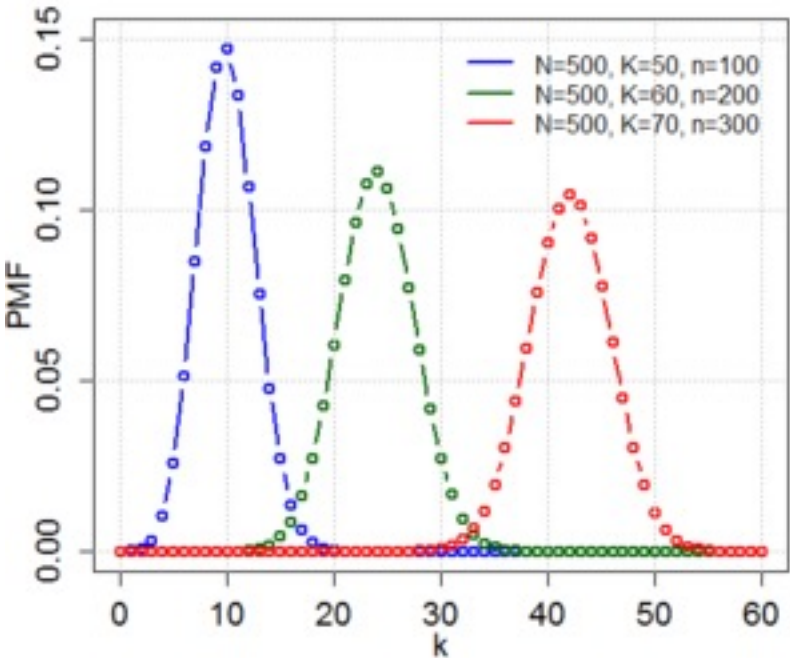
scRNA-seq based
single cell type



Liver
Tissue

Hypergeometric test used for enrichment analysis

Sampling without replacement



Expected

	in list	not in list	totals
with annotation	$(A+B)(A+C)/N$	$(A+B)(B+D)/N$	A+B
without annotation	$(C+D)(A+C)/N$	$(C+D)(B+D)/N$	C+D
	A+C	B+D	N

The following conditions characterize the hypergeometric distribution:

- The result of each draw (the elements of the population being sampled) can be classified into one of **two mutually exclusive categories** (e.g. Pass/Fail or Employed/Unemployed).
- The probability of a success changes on each draw, as each draw decreases the population (*sampling without replacement* from a finite population).

A **random variable** X follows the hypergeometric distribution if its **probability mass function** (pmf) is given by^[1]

$$p_X(k) = \Pr(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}},$$

where

- N is the population size,
- K is the number of success states in the population,
- n is the number of draws (i.e. quantity drawn in each trial),
- k is the number of observed successes,
- $\binom{a}{b}$ is a **binomial coefficient**.

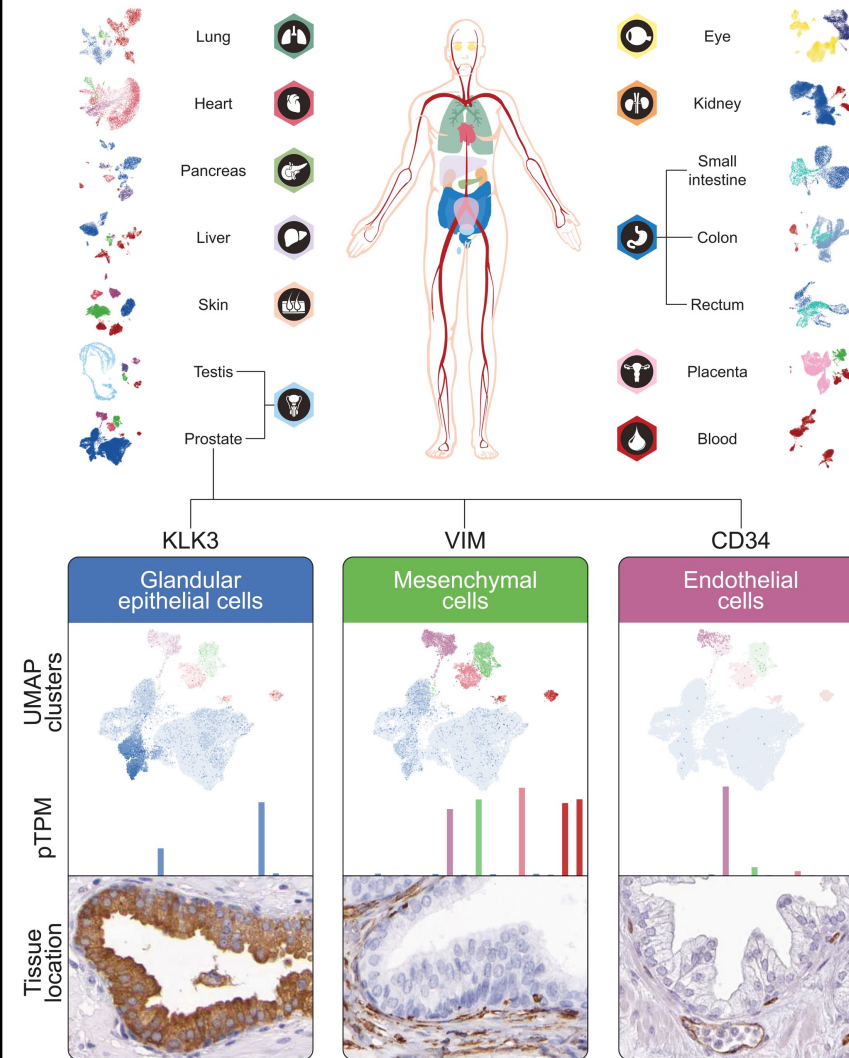
The pmf is positive when $\max(0, n + K - N) \leq k \leq \min(K, n)$.

A random variable distributed hypergeometrically with parameters N , K and n is written $X \sim \text{Hypergeometric}(N, K, n)$ and has **probability mass function** $p_X(k)$ above.

Atlas exploration:

- HUMAN PROTEIN ATLAS>SINGLE CELL TYPE

<https://www.proteinatlas.org/humanproteome/single+cell+type>



Discussion

- **Potential issues:** not reporting detailed procedures to select them, especially the name or number of papers in- or excluded
- **Applications:** gene expression profile $\neq$ protein profile $\neq$ biological functions (but we define them quite arbitrarily)
 - Redefinition of cell types
 - Optimization of cell type biomarkers
- **Future directions:** development of a temporal single cell transcriptomics profiling (mixed > where > when & where?)

Discussion

- **Cellular deconvolution:**
 - measure cell type composition, especially the changes of cell types by comparing to a reference (like in this study) to infer aetiology
- **Temporal profiling of transcriptomics:**
 - Sampling is difficult, as at any given time, there may be only very few cells at the desired stage, especially in adult humans
 - Cells at different stages requires different biomarkers. Such biomarkers haven't been available to all cell types due to limited knowledge in cellular states/stages
 - Difficult to mark and understand noncoding gene expression changes



Q&A

Presenters: Yaoding Chen, Ge Qu

References

- Franssen CL, Bardi M, Lambert KG. Using a comparative species approach to investigate the neurobiology of paternal responses. *JoVE (Journal of Visualized Experiments)*. 2011 Sep 19(55):e3173.
- Jaworski A, Tom I, Tong RK, Gildea HK, Koch AW, Gonzalez LC, Tessier-Lavigne M. Operational redundancy in axon guidance through the multifunctional receptor Robo3 and its ligand NELL2. *Science*. 2015 Nov 20;350(6263):961-5.
- Kiselev VY, Andrews TS, Hemberg M. Challenges in unsupervised clustering of single-cell RNA-seq data. *Nature Reviews Genetics*. 2019 May;20(5):273-82.
- Liu Z. Clustering Single-Cell RNA-Seq Data with Regularized Gaussian Graphical Model. *Genes*. 2021 Feb;12(2):311.
- Steinheuer LM, Canzler S, Hackermüller J. Benchmarking scRNA-seq imputation tools with respect to network inference highlights deficits in performance at high levels of sparsity. *BioRxiv*. 2021 Jan 1.